

Discrete Platinum(II) Metallacycles with Inner- and Outer-Modified 9,10-Distyrylanthracene: Design, Self-Assembly, and Luminescence Properties

Hao Yu, Junjuan Shi, Meng Li, Guocui Pan, Hui Tong, Bin Xu,* Ming Wang,* and Wenjing Tian



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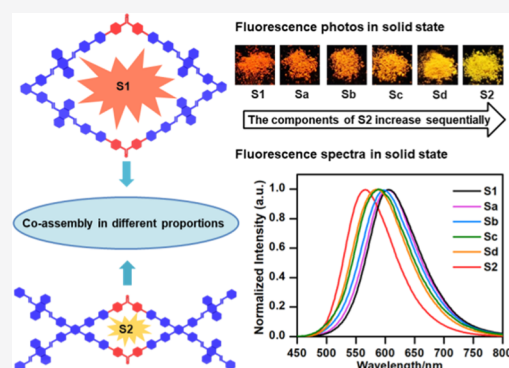


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Supporting Information

ABSTRACT: To completely unravel the relationship between structures and luminescence properties of inner- and outer-modified metallacycles, two rhombic metallacycles **S1** and **S2** with 9,10-distyrylanthracene (DSA) were constructed herein via the design of DSA moieties on the inside or outside of metallacycles. Similar building blocks of the two metallacycles led to the same emission wavelengths in a dilute solution. In contrast, their fluorescence emissions in the aggregation and solid states were significantly different and revealed interesting emission behaviors from structures with inner- and outer-modified design. Finally, based on their luminescence properties, a tunable solid-state fluorescence emissive material was easily obtained by the mixing of two supramolecules in different ratios. This unique design suggested that the modified position of fluorophores exerts a key impact on the adjustment of luminescence from a dilute solution to the aggregated state and will be of great significance for the development of luminescent materials based on supramolecules.



INTRODUCTION

Luminescent materials play a key role in the development and progress of human society and have attracted considerable attention owing to their widespread applications, such as light-emitting diodes, fluorescent probes, bioimaging, and so forth.¹ Among all luminescent materials, discrete supramolecular architectures based on self-assembly with noncovalent interactions (such as hydrogen bonds, metal–ligand bonds, and π – π interactions)² have become a fairly effective approach to investigate luminescence properties due to unique characteristics of discrete supramolecules, including precise shape, size, stoichiometry, and an accurate position of different components.³ Among them, coordination-driven self-assembly based on metal–ligand bonds is a powerful assembly strategy, with a high directional force and diversity of organic ligands,⁴ and it is typically utilized to construct discrete luminescent supramolecular architectures.⁵

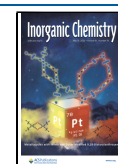
To overcome aggregation-caused quenching of a majority of fluorescent dyes in the aggregated state and obtain a high luminescence efficiency,⁶ in 2001, Tang et al. pioneered aggregation-induced emission (AIE),⁷ and AIE materials have attracted worldwide attention and research since then. Although a large number of emissive supramolecular coordination complexes (SCCs) based on AIE-related motifs have been reported including classic tetraphenylethylene^{4c,5a–c,8} and 9,10-distyrylanthracene (DSA),^{5h} chromophores with AIE activity mainly render a specific geometry, and almost all of the chromophores are placed on the contour

of discrete structures. Modification of AIE fluorophores on the inside and outside of SCCs to investigate the conformational effect for luminescence properties has rarely been reported,⁹ especially for discrete two-dimensional (2D) metallacycles. Considering that planar 2D supramolecules with rigid aromatic backbones exhibit strong π – π interactions, which in turn can lead to significant changes in terms of luminescence properties in different states,^{4c,5f,10} the exploration of the relationship between 2D metallacycles with different modification positions and fluorescence properties is more crucial for the construction of functional supramolecular architectures.

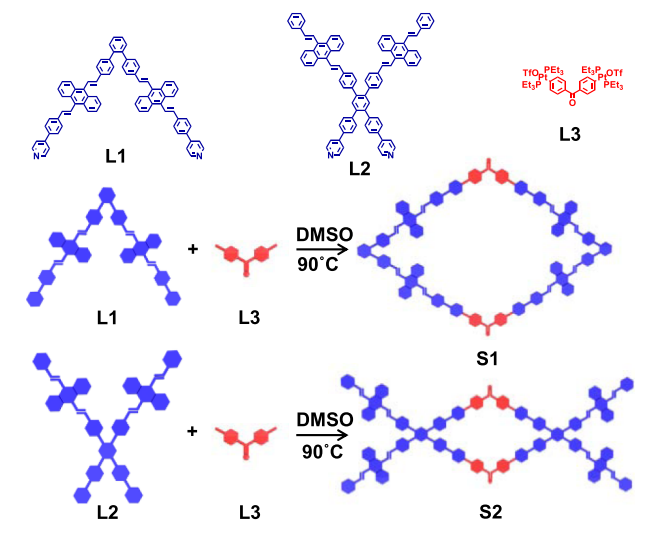
In this study, two inner- and outer-modified rhombic metallacycles were designed and constructed by employing a coordination-driven self-assembly strategy. “V”-type ligand **L1** and “X”-type ligand **L2**, both of which exhibited a molecular angle of 60° and with DSA groups on the inside and outside of ligands, respectively, were designed (Scheme 1). These two electron-rich pyridyl-based ligands can assemble with an electron-deficient di-Pt(II) acceptor **L3** at 120° to form rhombus metallacycles **S1** and **S2**. In addition, photophysical properties of **S1** and **S2** in a dilute solution and in the

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Scheme 1. Self-Assembly of Metallacycles S1 and S2



aggregated state were investigated. Distinct luminescence properties in different aggregated states were noted. Based on their luminescence characteristics, solid-state luminescent materials with tunable emission were successfully obtained by the mixing of these two complexes at different stoichiometric ratios.

EXPERIMENTAL SECTION

Ligand L3 was prepared according to the reported literature.¹¹ Nuclear magnetic resonance (NMR) spectral data were recorded on Bruker AVANCE 500 MHz, 600 MHz NMR, and Qone AS 400 MHz NMR spectrometers in CDCl₃ and dimethyl sulfoxide (DMSO-*d*₆). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was performed on a Bruker AutoflexIII using *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyldiene]malononitrile as a matrix. Electrospray ionization (ESI)-MS was performed using a Waters Synapt G2 tandem mass spectrometer. The UV-vis spectra of solution samples were recorded using a Shimadzu UV2550 spectrophotometer. The emission spectra of solution samples were measured on a Shimadzu RF-5301 PC spectrometer (CCD) and a Maya2000Pro optical fiber spectrophotometer. The UV-vis spectra of solid samples were recorded using a PerkinElmer Lambda950 spectrophotometer. The emission spectra of solid samples and solid-state quantum yields were determined using an Edinburgh FLS920 steady-state and transient state fluorescence spectrometer.

Synthesis of Ligand L1. Compound 7 (0.26 g, 0.5 mmol), compound 9 (0.066 g, 0.23 mmol), and potassium *tert*-butoxide (0.23 g, 2 mmol) were added to a 100 mL Schlenk bottle. The Schlenk bottle was evacuated under vacuum and flushed with nitrogen three times, and then, 30 mL of tetrahydrofuran (THF) was added. The mixture was reacted at room temperature for 6 h. After the reaction, the reaction solution was poured into 50 mL of methanol. After leaving the solution standing for a period of time, a yellow solid could be observed to precipitate from the methanol. Next, the precipitate was filtered and rinsed twice with methanol to obtain the final ligand L1 (0.16 g, 70%). ¹H NMR (500 MHz, CDCl₃, 300 K) δ (ppm): 8.73–8.68 (m, 4H, Ph-H^a), 8.44 (dd, *J* = 6.8, 3.2 Hz, 4H, Ph-H^b), 8.40 (dd, *J* = 6.7, 3.4 Hz, 4H, Ph-H^c), 8.04 (d, *J* = 16.7 Hz, 2H, Ph-H^d), 7.98 (d, *J* = 16.6 Hz, 2H, Ph-H^e), 7.81 (d, *J* = 8.2 Hz, 4H, Ph-H^f), 7.76 (d, *J* = 8.3 Hz, 4H, Ph-H^g), 7.66 (d, *J* = 8.2 Hz, 4H, Ph-H^h), 7.61–7.57 (m, 4H, Ph-Hⁱ), 7.57–7.46 (m, 12H, Ph-H^{j,k,l}), 7.37 (d, *J* = 8.1 Hz, 4H, Ph-H^m), 7.00 (d, *J* = 12.1 Hz, 2H, Ph-Hⁿ), 6.96 (d, *J* = 12.3 Hz, 2H, Ph-H^o). ¹³C NMR (125 MHz, CDCl₃, 300 K): δ 150.52, 147.91, 141.58, 140.23, 138.29, 137.72, 137.42, 136.74, 135.73, 133.87, 133.17, 132.43, 131.00, 130.58, 129.75, 127.93, 127.60,

127.47, 126.73, 126.59, 126.47, 125.57, 125.47, 125.14, 121.55. MALDI-TOF MS (*m/z*): calcd for [C₇₆H₅₂N₂-e]⁺, 992.41; found, 992.41.

Synthesis of Ligand L2. Compound 10 (0.25 g, 0.58 mmol), compound 16 (0.115 g, 0.19 mmol), and potassium *tert*-butoxide (0.23 g, 2.4 mmol) were added to a 100 mL Schlenk bottle. Then, 30 mL of THF was added under N₂. The reaction temperature was room temperature, and the reaction time was 6 h. After the reaction, the reactants were added to 50 mL of methanol. After leaving the solution standing for 10 min, a yellow solid could be seen to precipitate out of the system. Next, the solid in the bottle was suction-filtered for collection and washing with methanol was performed to obtain the final ligand L2 (0.15 g, 69%). ¹H NMR (500 MHz, CDCl₃, 300 K) δ (ppm): 8.69–8.64 (m, 4H, Ph-H^a), 8.46–8.37 (m, 8H, Ph-H^b), 8.00 (d, *J* = 16.6 Hz, 2H, Ph-H^c), 7.94 (d, *J* = 16.6 Hz, 2H, Ph-H^d), 7.70 (d, *J* = 7.8 Hz, 10H, Ph-H^{e,f}), 7.62 (d, *J* = 8.1 Hz, 4H, Ph-H^g), 7.57–7.52 (m, 4H, Ph-H^h), 7.47 (tt, *J* = 8.4, 5.1 Hz, 20H, Ph-H^{i,j,k,l}), 7.37 (t, *J* = 7.4 Hz, 2H, Ph-H^m), 6.96 (t, *J* = 17.2 Hz, 4H, Ph-Hⁿ). ¹³C NMR (125 MHz, CDCl₃, 300 K): δ 150.41, 147.82, 141.77, 141.14, 140.69, 139.90, 139.75, 139.24, 139.15, 137.67, 137.42, 137.12, 136.66, 136.42, 136.28, 135.98, 135.12, 133.32, 130.80, 130.49, 129.73, 129.00, 128.80, 128.22, 127.75, 126.97, 126.75, 126.67, 126.56, 125.43, 125.28, 122.64, 122.58, 121.55, 119.79, 119.64. MALDI-TOF MS (*m/z*): calcd for [C₇₆H₅₂N₂-e]⁺, 1144.48; found, 1144.48.

Synthesis of Metallacycle S1. Ligand L1 (1.98 mg, 0.002 mmol) and ligand L3 (2.68 mg, 0.002 mmol) were added in accurate ratios, and then, 1 mL of DMSO was added. The mixture was heated to 90 °C for 10 h. After the assembly was completed, the assembly solution was dropped into 8 mL of ether. After leaving the solution standing for 10 min, the precipitated solid was centrifuged and washed twice with ether. Finally, an orange-red solid was obtained (4.19 mg, 90%). ¹H NMR (500 MHz, DMSO-*d*₆) δ 8.85 (s, 8H, Ph-H^a), 8.54–8.29 (m, 20H, Ph-H^{b,c,d}), 8.24–8.00 (m, 28H, Ph-H^{e,f,g,h}), 7.81 (d, *J* = 7.7 Hz, 8H, Ph-Hⁱ), 7.72–7.50 (m, 32H, Ph-H^{j,k,l,m,n}), 7.48 (d, *J* = 10.6 Hz, 8H, Ph-H^o), 7.41–7.26 (m, 8H, Ph-H^p), 7.09 (d, *J* = 16.4 Hz, 4H, Ph-H^q), 6.97 (dd, *J* = 16.4, 7.9 Hz, 4H, Ph-H^r), 1.44–1.31 (m, 48H, Ph-H^{s,t}), 1.10 (q, *J* = 8.2 Hz, 72H, Ph-H^{u,v}). ¹³C NMR (125 MHz, DMSO-*d*₆, 300 K): δ 152.46, 148.79, 143.22, 140.99, 139.53, 139.31, 136.92, 136.24, 136.05, 135.27, 133.89, 132.71, 132.31, 131.90, 130.04, 128.95, 128.80, 127.82, 127.61, 127.08, 126.74, 125.77, 125.69, 124.84, 123.98, 11.99, 7.37. ESI-MS (*m/z*): 1017.6 [M-4OTf]⁴⁺ (calcd *m/z*: 1017.6).

Synthesis of Metallacycle S2. Ligand L2 (2.29 mg, 0.002 mmol) and ligand L3 (2.68 mg, 0.002 mmol) were added in accurate ratios, and then, 1 mL of DMSO was added. The mixture was heated to 90 °C for 10 h. After the assembly was completed, the assembly solution was dropped into 8 mL of ether. After leaving the solution standing for a period of time, a yellow precipitate would be obtained. The precipitate was centrifuged and washed with ether again. Finally, a yellow solid was obtained (4.52 mg, 91%). ¹H NMR (500 MHz, DMSO-*d*₆, 300 K): δ (ppm): 8.83 (s, 8H, Ph-H^a), 8.42 (ddt, *J* = 12.5, 6.0, 3.0 Hz, 16H, Ph-H^b), 8.24 (d, *J* = 16.9 Hz, 4H, Ph-H^c), 8.19–8.09 (m, 12H, Ph-H^d), 8.03 (d, *J* = 8.6 Hz, 8H, Ph-H^e), 7.89 (d, *J* = 8.0 Hz, 8H, Ph-H^f), 7.84 (d, *J* = 7.7 Hz, 8H, Ph-H^g), 7.72 (s, 4H, Ph-H^h), 7.69–7.54 (m, 24H, Ph-H^{i,j,k}), 7.54–7.44 (m, 16H, Ph-H^l), 7.40 (d, *J* = 7.2 Hz, 4H, Ph-H^m), 7.02 (d, *J* = 16.3 Hz, 4H, Ph-Hⁿ), 6.95 (dd, *J* = 16.4, 2.5 Hz, 4H, Ph-H^o). ¹³C NMR (125 MHz, DMSO-*d*₆, 300 K): δ 152.46, 148.72, 148.57, 143.23, 142.83, 142.54, 139.89, 139.50, 138.37, 138.18, 137.26, 136.83, 136.73, 135.97, 135.77, 132.57, 132.40, 132.22, 130.86, 130.08, 128.94, 128.79, 128.16, 127.14, 126.88, 126.76, 126.22, 125.63, 125.22, 124.77, 124.19, 123.97, 11.96, 7.35. ESI-MS (*m/z*): 1508.2 [M-3OTf]³⁺ (calcd *m/z*: 1508.2).

RESULTS AND DISCUSSION

Two 60° ditopic pyridyl-based ligands L1 and L2 were designed and synthesized by Suzuki coupling reactions and Wittig–Horner reactions. Supporting Information Schemes S1

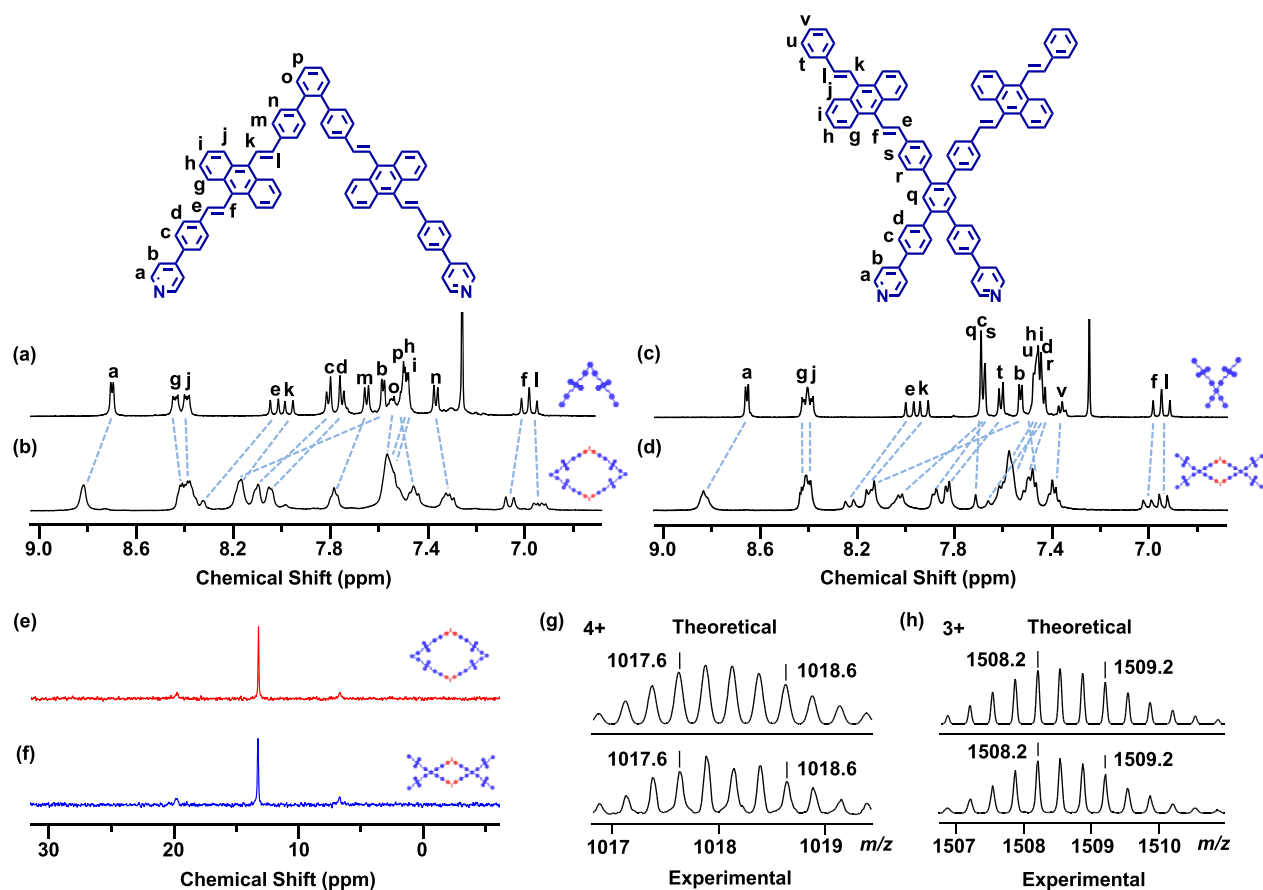


Figure 1. ^1H NMR spectra (500 MHz, 300 K) of **L1** (a) and **L2** (c) in CDCl_3 and **S1** (b) and **S2** (d) in $\text{DMSO}-d_6$. $^{31}\text{P}\{^1\text{H}\}$ NMR spectra (202 MHz, 300 K) of **S1** (e) and **S2** (f) in $\text{DMSO}-d_6$. ESI-MS spectra of **S1** (g) and **S2** (h).

and **S2** show the detailed synthetic routes. The two metallacycles (**S1** and **S2**) were assembled by mixing the donor ligands **L1** or **L2** with a 120° diplatinum(II) acceptor **L3** in a 1:1 stoichiometric ratio in DMSO at 90°C . After 10 h, the final discrete metallacycles were obtained in good yields (90% for **S1** and 91% for **S2**) by sedimentation and centrifugation.

These two ligands and complexes were completely characterized using a series of spectroscopic techniques, including multinuclear NMR [^1H , ^{13}C , $^{31}\text{P}\{^1\text{H}\}$, 2D correlation spectroscopy, 2D nuclear Overhauser effect spectroscopy, and 2D diffusion-ordered NMR spectroscopy (DOSY)] (Figures S16–S35) and MALDI-TOF-MS (Figures S39 and S40). ^1H NMR and $^{31}\text{P}\{^1\text{H}\}$ NMR patterns of these two complexes revealed only one set of signals, indicating the formation of a discrete assembly with high symmetry (Figure 1). Compared to the proton signals of ligands (Figure 1a,c), the α and β hydrogen atoms (H^a and H^b) of the pyridine rings in metallacycles **S1** and **S2** exhibited significant downfield shifts (Figure 1b,d) due to the loss of electron density after complexation with the $\text{Pt}(\text{II})$ acceptors.¹² $^{31}\text{P}\{^1\text{H}\}$ NMR spectra of these two complexes revealed sharp singlets with concomitant ^{195}Pt satellites, corresponding to a single phosphorus environment. Moreover, owing to the similar building blocks of these two assemblies, $^{31}\text{P}\{^1\text{H}\}$ NMR spectra of **S1** and **S2** only exhibited slightly different values (ca. 13.30 ppm for **S1** and 13.25 ppm for **S2**) (Figure 1e,f).^{12c,13} DOSY was employed to measure the sizes of these two complexes and exhibited distinct bands at $\log D = -10.47$ for **S1** and $\log D =$

-10.33 for **S2**, indicative of the formation of a single product. Based on the Stokes–Einstein equation,¹⁴ the diameters of **S1** and **S2** were calculated to be 5.76 and 4.21 nm, respectively. The experimental diameter of macrocycle **S2** was less than that of **S1**, possibly related to the high degree of freedom of the DSA group in **S2**.

ESI-MS was further employed to determine the molecular compositions of these two assemblies. For **S1**, an isotopically well-resolved peak at $m/z = 1017.6$ (Figure 1g) was observed, corresponding to the charge state 4+ of metallacycle **S1**. The experimental isotope pattern matched well with the corresponding simulated isotope pattern. After the deconvolution of m/z , the calculated molecular weight of **S1** was 4666.2 Da, which was in good agreement with the formula of $[\text{C}_{230}\text{H}_{240}\text{F}_{12}\text{N}_4\text{O}_{14}\text{P}_8\text{Pt}_4\text{S}_4]$. Following the same calculation, the molecular weight of **S2** was estimated to be 4971.15 Da, which was consistent with the expected formula of $[\text{C}_{254}\text{H}_{256}\text{F}_{12}\text{N}_4\text{O}_{14}\text{P}_8\text{Pt}_4\text{S}_4]$. All these characterization results suggested that two rhombic metallacycles with two pyridyl-based ligands and two diplatinum(II) acceptors were successfully prepared.

After completely determining the structural compositions of metallacycles **S1** and **S2**, photophysical properties of all ligands and complexes ($c = 1.00\ \mu\text{M}$) were further investigated using UV-vis spectroscopy and fluorescence spectroscopy for clarifying the effect of functional group DSA at different positions in the complexes. As shown in Figure 2, **L1** and **L2** exhibited similar absorption bands at 270, 320, and 420 nm, corresponding to the same building blocks of the benzene/

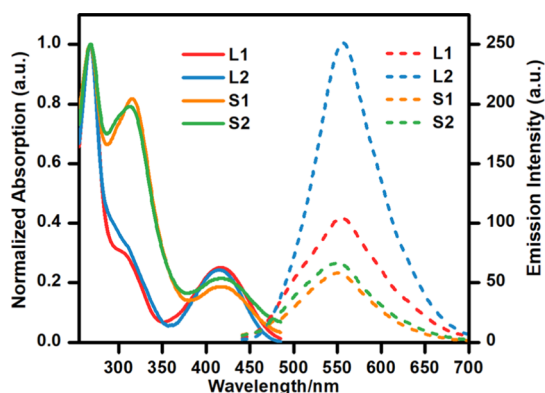


Figure 2. UV-vis spectra and fluorescence emission spectra of L1 and L2 in CHCl_3 and S1 and S2 in DMSO ($\lambda_{\text{exc}} = 420 \text{ nm}$, $c = 1.00 \mu\text{M}$).

pyridine segment, the styrene segment, and the $\pi-\pi^*$ transition, respectively.¹⁵ Under the excitation of a 420 nm light source, the two ligands also exhibited the same emission bands at 556 nm. After coordination self-assembly with L3, metallacycles S1 and S2 exhibited three absorption bands similar to that of L1 and L2. Fluorescence spectra of these two metallacycles revealed a slight blue shift and reduced emission intensity after complexation, and S2 still exhibited a relatively high emission intensity. However, the emission bands of these two complexes were also still the same at 551 nm, suggesting that the positions of the DSA group did not affect the fluorescence emission band in a dilute solution.

Next, the photophysical behaviors in the aggregated state of two metallacycles modified with DSA moieties in different positions were further investigated. Fluorescence spectra of metallacycles S1 and S2 in DMSO with different H_2O fractions were recorded (Figure 3a,b). With the increase in the fraction of H_2O as a poor solvent from 0 to 90%, the emission

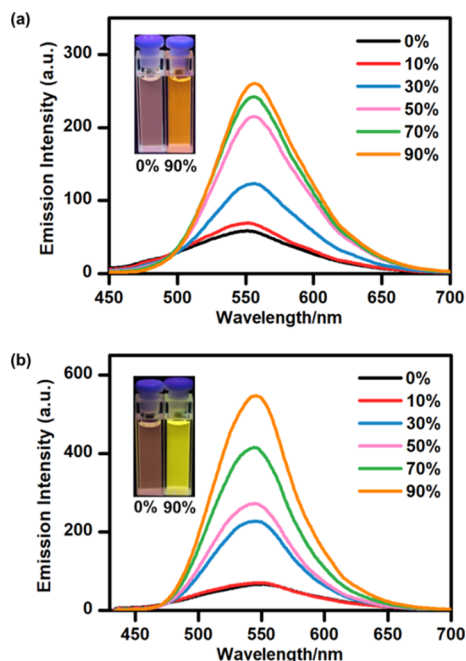


Figure 3. Fluorescence emission spectra of S1 (a) and S2 (b) vs H_2O fractions in DMSO/ H_2O mixtures ($\lambda_{\text{exc}} = 420 \text{ nm}$, $c = 1.00 \mu\text{M}$).

intensities of the two metallacycles gradually increased, indicative of typical AIE properties. However, notably, the addition of H_2O led to a shift in the emission bands of the two assemblies in different directions. For S2, the emission band was blue-shifted from 551 to 545 nm, showing a shift change just opposite to that of S1 (from 551 to 556 nm). Correspondingly, with the gradual addition of water, intermolecular accumulation increased, and the colors of S1 and S2 solutions gradually changed to orange-red and yellow, respectively. Typically, the emission band shift caused by adding poor solvents is affected by charge transfer (CT) in the molecule, and the CT process leads to the red shift of the emission band for the AIE fluorophore in a polar solution.¹⁶ As a result, the addition of polar solvent H_2O caused the red shift in the emission band of metallacycle S1. However, this explanation was clearly not applicable to S2. For S2, four DSA moieties were distributed on the outside of the rhombic skeleton and exhibited a high degree of freedom. It is hypothesized that by the addition of H_2O , the individual assembly gathers together to form microaggregates, thereby providing an internal hydrophobic environment to reduce the CT effect and subsequently resulting in the blue shift of the emission band of S2.^{16,17} In addition, we also explored the electronic transitions in the excited states of ligands and metallacycles through time-dependent density functional theory (TD-DFT) calculations (Figures S50–S52). Tables S1–S3 and Figure S45 show the relative fluorescence quantum yields (Φ_F) of the two metallacycles with different H_2O contents. Relative Φ_F values for metallacycles S1 and S2 in pure DMSO were 7 and 15%, respectively. With the increasing water fraction, the rotation of DSA motifs was gradually limited with molecular aggregation, and the relative quantum yields of S1 and S2 in the aggregated state (90% water fraction) increased to 16 and 21%, respectively. Although the fluorescence intensity of S2 was still stronger than that of S1, the relative Φ_F value of S2 was only increased by 1.4 times. However, the Φ_F value of S1 at 90% H_2O increased by 2.3 times compared to that in pure DMSO, suggesting that S1 with the inner-modified DSA motifs exhibited a more obvious AIE effect. Compared with S2 containing DSA moieties on the outside, S1 exhibited four DSA moieties on the inside of the structure, rendering clear advantages for the accumulation and restriction of DSA torsion during aggregation. These results suggested that the distributions of luminescent groups exert a key impact on the performance of coordinated supramolecules. A reasonable design of the position of luminescent groups can achieve the precise adjustment of the luminescence properties of supramolecules.

The studies on different photophysical properties of metallacycles S1 and S2 in the solution and aggregated states attracted our interest in examining their solid-state luminescence properties. Solids L1, L2, and S2 were yellow powders under ambient light, but S1 was an orange-red powder (Figure 4c). The absorption band of S1 was significantly red-shifted and broadened in comparison with those of L1, L2, and S2 (Figure 4a), suggesting that S1 exhibits a larger π -conjugated system and aggregation degree in the solid state.^{10b} Under a 365 nm UV light, L1, L2, and S2 exhibited yellow fluorescence, and S1 exhibited orange-red fluorescence. The fluorescence colors of solids S1 and S2 were consistent with fluorescence in the case of the introduction of 90% H_2O into their dilute solution. Solid-state fluorescence quantum yields of the four compounds L1, L2, S1, and S2 were 5, 27, 17, and

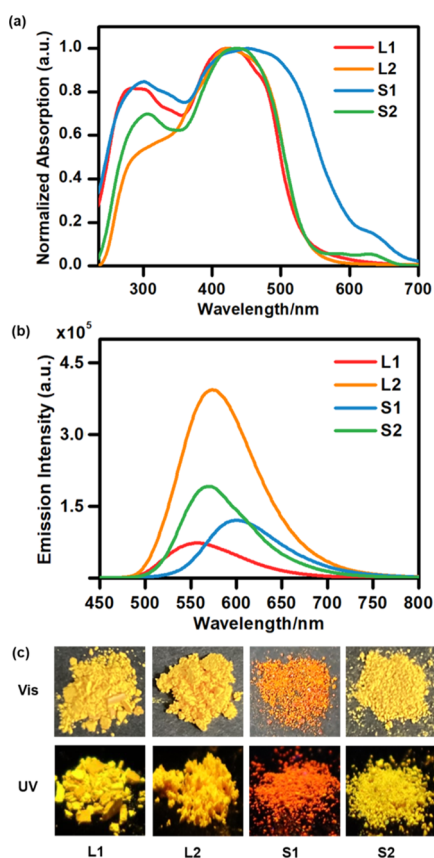


Figure 4. (a) UV-vis spectra and (b) fluorescence spectra of L1, L2, S1, and S2 in the solid state ($\lambda_{\text{ex}} = 420$ nm). (c) Photographs of L1, L2, S1, and S2 in the solid state. Up: under ambient light and down: under a UV lamp at 365 nm.

24%, respectively. In the solid state, ligand L2 (573 nm) and metallacycle S2 (570 nm) exhibited similar emission bands, which complied with their liquid fluorescence emission characteristics. However, metallacycle S1 exhibited a completely different fluorescence emission behavior. Compared with the emission of S1 in a dilute solution at 551 nm, complex S1 (601 nm) in the solid state exhibited a large red shift of 50 nm (Figure 4b). In complex S1, owing to the restriction of DSA moieties, the locked DSA group was planarized, which was extremely advantageous for intermolecular aggregation and stacking, leading to the further red shift of emission.^{5h}

Based on the study of the fluorescence properties of metallacycles S1 and S2, we speculated that by mixing the two metallacycles in different proportions, the solid-state fluorescence of the mixtures could be tuned within a specified range. We mixed the solutions of S1 and S2 at the molar ratio of 1:1 and left them at room temperature for 1 day. The individual signals from S1 and S2 were not clearly distinguished from the ¹H and ³¹P{¹H} NMR spectra of the mixture possibly due to the presence of similar building blocks and chemical environment (Figures S36 and S37). However, the ESI-MS result revealed distinguishable S1 and S2 (Figures S43 and S44), and the 2D DOSY spectrum of the mixture revealed two bands of S1 and S2 (Figure S38), suggesting that there was no ligand exchange between the two metallacycles. On the basis of this, the solution of S1 and S2 was mixed in six ratios, namely, 10:0 (S1), 8:2 (Sa), 6:4 (Sb), 4:6 (Sc), 2:8 (Sd), and 0:10 (S2), and the fluorescence emissions of their solution and solid states were investigated. With the increase of

the S1 composition, the lambda of six substances in the absorption bands of 270 and 320 nm gradually decreased (Figure S48). However, due to the similar fluorescence emission of the two metallacycles, the fluorescence of these six substances in a dilute solution did not exhibit any clear difference (Figure 5a). Subsequently, we precipitated the six

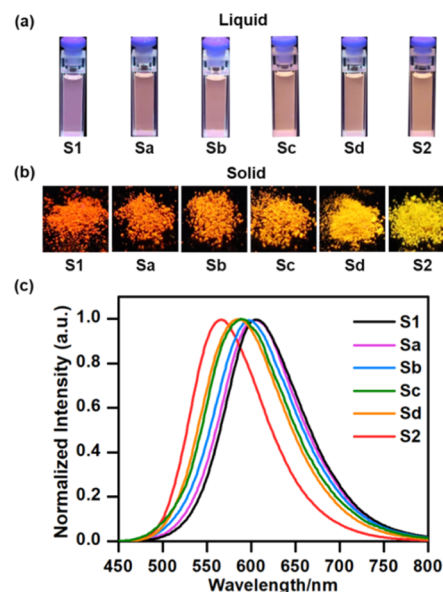


Figure 5. (a) Photographs of S1, Sa, Sb, Sc, Sd, and S2 in DMSO ($c = 1.00 \mu\text{M}$) under a UV lamp at 365 nm. (b) Photographs of S1, Sa, Sb, Sc, Sd, and S2 in the solid state under a UV lamp at 365 nm. (c) Fluorescence spectra of S1, Sa, Sb, Sc, Sd, and S2 in the solid state ($\lambda_{\text{ex}} = 420$ nm).

substances with ether, and after centrifuging and drying, the solids of the six substances were obtained. In sharp contrast to the case of the solution state, the solid-state fluorescence emissions of these six substances were significantly different, with a gradual change from orange-red to yellow; hence, solid-state fluorescence within a certain range can be tuned (Figure 5b). This gradation of fluorescence colors can be better reflected in the solid-state fluorescence spectrum (Figures 5c and S49). The emission bands of the six substances gradually blue-shifted from S1 to S2, just corresponding to the gradation of fluorescence colors.

CONCLUSIONS

In summary, based on inner- and outer-modified DSA moieties, two rhombus metallacycles S1 and S2 were designed and constructed via coordination-driven self-assembly. In the solution state ($c = 1.00 \mu\text{M}$), complexes S1 and S2 exhibited similar absorption and emission spectra, indicating that the positions of the fluorescent group did not significantly affect the fluorescence emission in a dilute solution. However, these two complexes exhibited clearly different luminescence behavior in the aggregated and solid states, corresponding to different internal and external modifications. Moreover, a series of tunable solid-state luminescent materials were successfully obtained by the facile mixing of these two complexes. Through the subtle design of these structures, the relationship between the structure and photophysical properties based on inner- and outer-modified metallacycles was successfully clarified, thereby

providing novel insights into strategies for the design and development of tunable luminescent materials.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.1c03433>.

Synthetic and characterization data obtained from NMR, ESI-MS, photophysical studies, and TD-DFT calculations (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Bin Xu – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China; orcid.org/0000-0002-0423-925X; Email: xubin@jlu.edu.cn

Ming Wang – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China; orcid.org/0000-0002-5332-0804; Email: mingwang358@jlu.edu.cn

Authors

Hao Yu – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China

Junjuan Shi – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China

Meng Li – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China

Guocui Pan – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China

Hui Tong – State Key Laboratory of Polymer Physics and Chemistry, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, China; orcid.org/0000-0001-7734-9532

Wenjing Tian – State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, P. R. China; orcid.org/0000-0001-5766-8050

Complete contact information is available at:

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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